Intensity, decibels and Doppler

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish intensity change from frequency shift.

SUCCESS CRITERIA - I can...

1. Use an intensity ratio separately from pitch.
2. Interpret a logarithmic sound-level difference.
3. Predict approach and recession Doppler trends.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “intensity” and “decibel” to explain your answer. Sentence starters are optional.

intensity: power passing through a unit area

decibel: a unit for a logarithmic sound-level comparison

Doppler shift: a frequency change associated with relative motion

01 Warm-up: bring back an earlier idea

A **500 Hz** sound travels through air. Do air particles mainly move parallel or perpendicular to propagation? **Name the wave type**.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

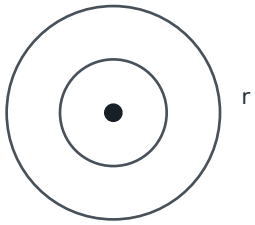
02 Worked example: explain the prediction

Reference: $\Delta\beta = \mathbf{10}\log_{10}(I_2/I_1)$.
I is intensity; β is sound level in dB. Tenfold intensity gives **+10 dB**. Approach generally raises observed frequency; recession lowers it.

Worked reasoning: For the same point source, doubling distance spreads its power over four times the area, so ideal intensity drops to one quarter. This assumes spherical spreading without extra absorption.

Your turn: Draw two concentric wavefronts and indicate why the outer surface shares the power more widely.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Label distances and compare intensity.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A moving source approaches a stationary listener, then recedes. **Sketch closer wavefront spacing** in front and wider spacing behind; **compare heard pitch**.

Sentence starter: Both ... . They differ in ... because ... .

Source moves right. Add wavefronts.



Marzano check - circle: ① ② ③

04 Apply the physics

A sound meter comparison uses the rule: multiplying intensity by **10** adds **10 dB**. **Complete the table** for intensity multipliers **10** and **100**. **Explain why 100** times intensity is not a **100 dB** increase.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

Case	Intensity ratio	Level change (dB)
A	____	____
B	____	____

Marzano check - circle: ① ② ③

Intensity, decibels and Doppler

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A loud stationary siren and a quiet approaching siren are compared. Does “louder” alone mean “higher frequency”? Explain **0**.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

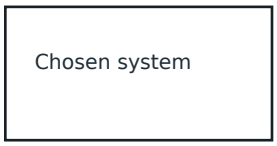
06 Review: connect an earlier idea

A sample warms by **4°C** after a certain energy input. If absorbed energy becomes **6** times as large with the same mass and material and no phase change, find the new rise.
Label energy entering, leaving and stored.

Word help: absorbed = taken into the specified object or system

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

An observer hears a steadily emitting siren approach. Is observed frequency higher or lower than emitted frequency?

Sentence starter: I notice … . This supports … because … .

Source moves right. Add wavefronts.



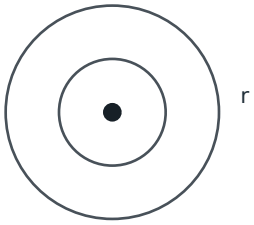
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Compare intensities at **1**, **2** and **3 m** from the same ideal source as fractions of the **1 m** value.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .



Label distances and compare intensity.

Marzano check - circle: 1 2 3